

Hierarchical Data Visualization

Comparative Analysis Report

Phase 1: Portfolio Checkpoint 2

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Executive Summary

This report presents a comparative study of hierarchical visualization techniques across three datasets: global population, company structure, and budget allocation.

Objective

To evaluate and compare the effectiveness of Treemaps, Dendrograms, Sunburst Charts, and Comparative Charts in uncovering hierarchical patterns within multi-level data.

Datasets Analyzed

- World Population (195 countries)
- Organizational Structure (500 employees)
- Budget Distribution (\$20M)

Methodologies

- Treemap Analysis
- Hierarchical Clustering
- Sunburst Visualization
- Comparative Assessment

1 Treemap Analysis

Treemaps were applied to visualize global population distribution across continents, with area representing population and color encoding GDP per capita.

Continent	Population Share
Asia	58.9%
Africa	18.1%
Europe	9.4%
North America	7.6%
South America	5.5%
Oceania	0.6%

Table 1: Population distribution by continent

- Demographic Concentration:** China and India dominate Asia's population.
- Economic Disparity:** GDP per capita differs greatly between continents.
- Wealth Distribution:** North America leads in GDP despite smaller population.

Strength: Treemaps clearly display proportional relationships while preserving hierarchy.

2 Hierarchical Clustering (Dendrogram)

Countries were grouped using the Ward method based on population and GDP per capita, forming five main clusters.

- 1. **Population Giants:** China, India
- 2. **High GDP Micro-states:** Monaco, Liechtenstein
- 3. **Developing Economies:** Brazil, Russia, Mexico, Indonesia, etc.
- 4. **Advanced Economies:** USA, Switzerland, Singapore, etc.
- 5. **Developed Nations:** Germany, Japan, UK, France, etc.

Strength: Dendrograms highlight similarity and grouping, ideal for comparative classification.

3 Sunburst Chart: Organizational Structure

A tech company’s hierarchy was visualized with a Sunburst chart, illustrating employee distribution.

Department	Employees	Share
Engineering	180	36%
Sales	120	24%
Operations	120	24%
Product	80	16%
Total	500	100%

Table 2: Employee distribution by department

- **Engineering Focus:** Core of company resources.
- **Balanced Allocation:** Sales and Operations evenly distributed.
- **Clear Hierarchy:** From company to team level.

Strength: Sunburst charts give an intuitive radial view of nested data structures.

4 Budget Distribution

Budget allocation was visualized using a circular treemap to display investments and priorities.

- **Tech Investment:** Half of total budget for engineering.

Department	Budget	Share	Avg/Project
Engineering	\$10M	50.0%	\$2.5M
Research	\$4M	20.0%	\$1.3M
Marketing	\$3.5M	17.5%	\$1.2M
Sales	\$2.5M	12.5%	\$0.8M

Table 3: Budget allocation by department

- **Innovation:** Research takes 20%, ensuring growth.
- **Resource Disparity:** Sales underfunded despite its role.

Strength: Circular treemaps reveal proportional budgets and priorities clearly.

5 Technique Comparison

A comparative assessment across all visualization methods was conducted.

Technique	Advantages	Disadvantages	Best For
Treemap	Compact and proportional	Hard to read deep hierarchies	Medium datasets
Dendrogram	Reveals structure	Lacks quantitative scale	Cluster analysis
Sunburst	Easy multilevel view	Labels overlap	Organizational data
Bar Chart	Simple comparison	No hierarchy	Small datasets

Table 4: Comparison of visualization techniques

6 Conclusions

Findings

1. Asia’s 58.9% population dominates, yet North America leads in GDP per capita.
2. The company analyzed invests heavily in engineering (50% budget).
3. Each visualization fits a specific purpose: Treemaps for proportions, Dendrograms for structure, Sunbursts for hierarchy.

Strengths

- Excellent for layered and big data.
- Adds color-coded insights.
- Efficient space use.

Limitations

- Hard for non-technical users.
- Label crowding in deep layers.
- Static formats lose interactivity.

Recommendations

1. Choose visualization by question, not aesthetics.
2. Use interactive versions for exploration.
3. Match complexity to audience knowledge.
4. Combine visuals for broader insight.

Summary

Hierarchical visualizations reveal patterns in complex datasets. Choosing the right method enhances both understanding and communication.

Key Takeaway

Selecting the right visualization type directly affects how insights are interpreted and communicated, making it a core skill for any data professional.

3

Datasets

4

Methods

5

Clusters

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